INCH-POUND
MIL-DTL-16377/8C(SH)
30 June 2014
SUPERSEDING
MIL-DTL-16377/8B(SH)
2 August 1996

DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; FLUORESCENT AND SSL EQUIVALENT, GENERAL LIGHTING, 20 WATTS, 120 VOLTS, 60 HERTZ, WATERTIGHT, SYMBOLS 331.1, 331.2, 347.2, 347.3, 331.1SSL, 331.2SSL, 347.2SSL, AND 347.3SSL

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

All dimensions are in inches. For tolerances other than specified herein, see 3.5.2 of MIL-DTL-16377.

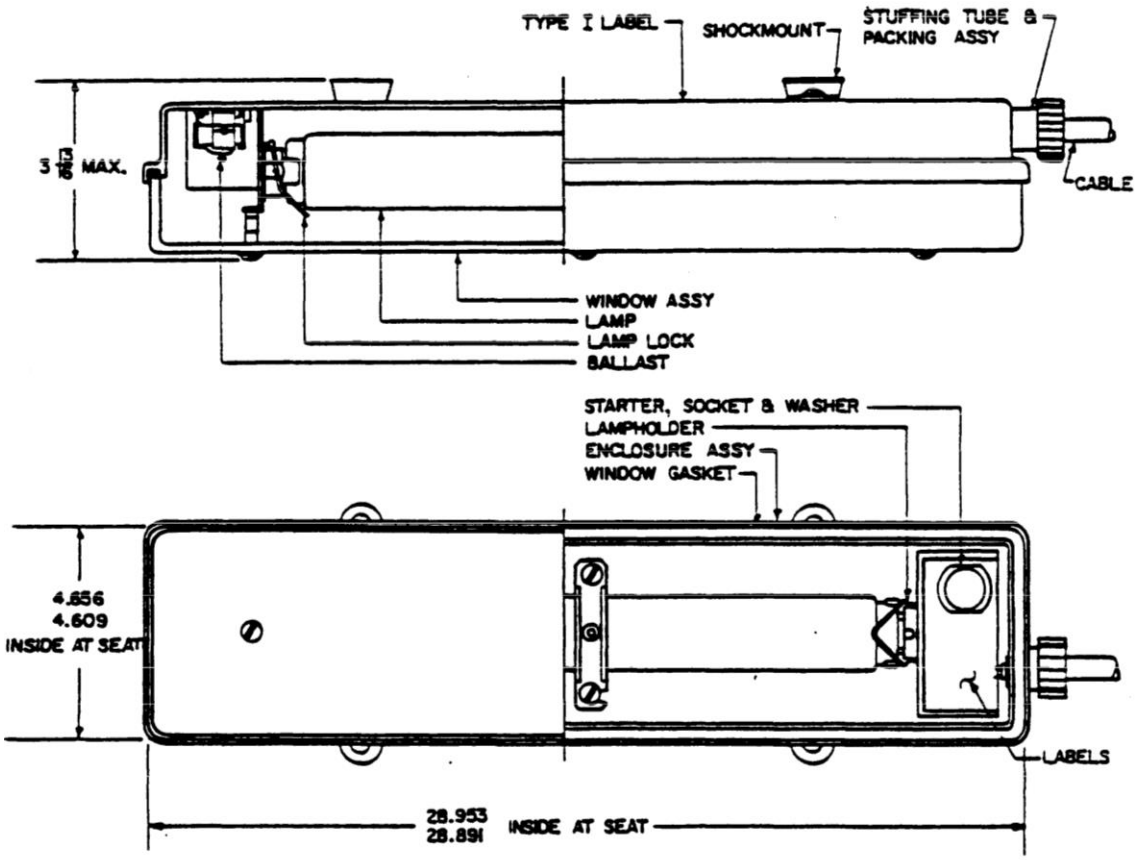


FIGURE 1. Typical view of symbols 331.1 and 347.2.

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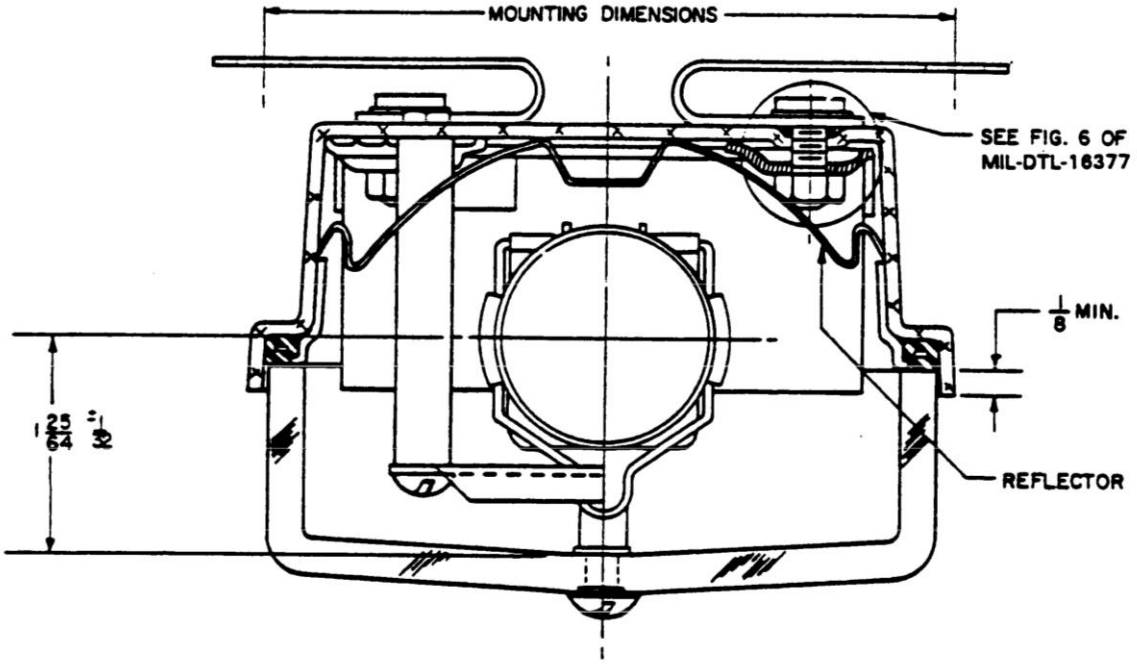


FIGURE 2. Typical section for symbols 331.1 and 347.2.

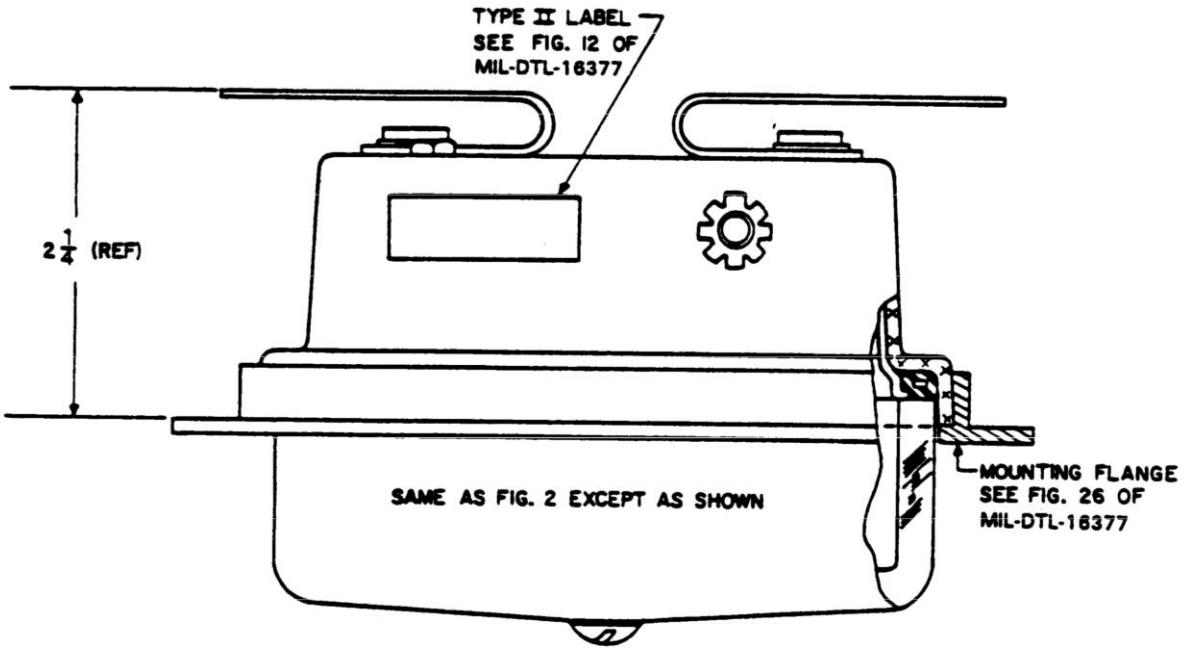
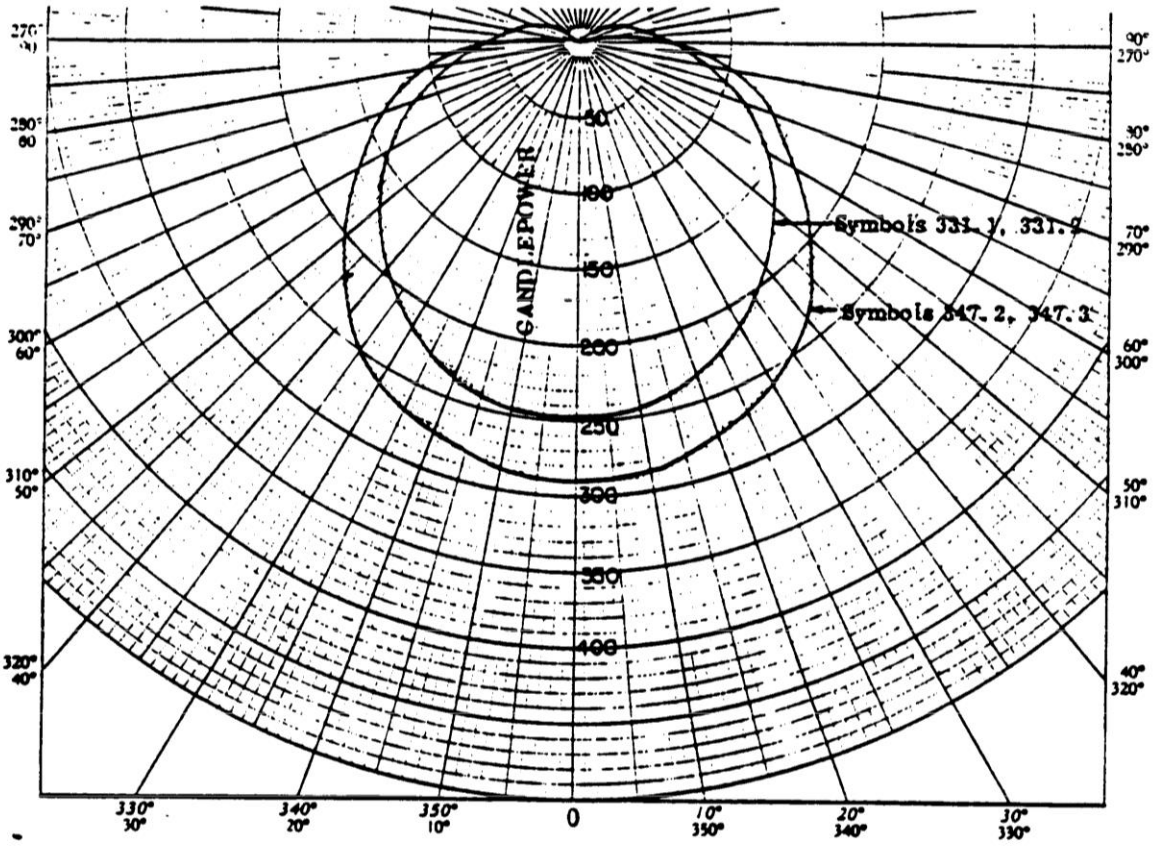


FIGURE 3. Typical section for symbols 331.2 and 347.3.

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PHOTOMETRIC TEST DATA:

Distance:	10 feet.
Plane:	At right angles to long axis of luminaire through geometric center of luminaire.
Voltage:	115 volts, 60 hertz.
Ambient temperature:	77 °F (25 °C).
Fluorescent lamps:	Use lamps furnished with fixtures seasoned for at least 100 hours.
Fluorescent ballast:	Use ballast furnished with fixture.
Fluorescent lamp correction:	Distribution curves based on 1,280 lumens per lamp.
SSL:	No seasoning required for LEDs.

FIGURE 4. Candlepower distribution curves.

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TABLE I. Numerical candela distribution for symbols 331.1, 331.2, 347.2, 347.3.

Angle	Candlepower	
	Symbol Number	
	331.1, 331.2	347.2, 347.3
0	243	285
5	242	285
10	240	282
15	238	278
20	230	270
25	225	260
30	216	255
35	205	250
40	194	235
45	182	218
50	170	200
55	160	185
60	145	168

Angle	Candlepower	
	Symbol Number	
	331.1, 331.2	347.2, 347.3
65	130	150
70	117	140
75	100	125
80	85	100
85	70	78
90	60	50
95	48	40
100	44	20
105	38	10
110	33	0
115	25	0
120	25	0
125	0	0

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FLUORESCENT REQUIREMENTS:

CLASSIFICATION:

Type: I
Class: 2

SYMBOL NUMBER:

331.1: Complete lighting fixture assembly, standard mounting, with one 20-watt cool white fluorescent lamp and white diffusing window assembly.
331.2: Complete lighting fixture assembly, flush mounting, with one 20-watt cool white fluorescent lamp and white diffusing window assembly.
347.2: Complete lighting fixture assembly, standard mounting, with one 20-watt cool white fluorescent lamp and clear and white prismatic window assembly.
347.3: Complete lighting fixture assembly, flush mounting, with one 20-watt cool white fluorescent lamp and clear and white prismatic window assembly.

DESIGN:

Electrical rating:	20 watts, 120 volts, 60 hertz.
Dimensions and configurations:	See figures 1, 2, and 3 and figure 10 of MIL-DTL-16377.
Photometric data:	See figure 4 and table I.
Lampholder spacing:	See figure 21 of MIL-DTL-16377.
Wiring diagram:	See figure 22 of MIL-DTL-16377.
Mounting dimensions:	
Symbols 331.1 and 347.2:	See figure 25 of MIL-DTL-16377.
Symbols 331.2 and 347.3:	See figure 26 of MIL-DTL-16377.
Window assembly:	
Symbols 331.1 and 331.2:	Military part number M16377/8-001, figure 27 of MIL-DTL-16377.
Symbols 347.2 and 347.3:	Military part number M16377/8-003, figure 27 of MIL-DTL-16377.
Enclosure:	Watertight. See 3.5.11 of MIL-DTL-16377.
Cable:	10 feet of Type LSDHOF-3 in accordance with MIL-DTL-24643/3.
Weight:	
Symbols 331.1 and 347.2:	8 pounds 10 ounces maximum.
Symbols 331.2 and 347.3:	10 pounds 2 ounces maximum.
Designation and marking:	The following labels shall be furnished in accordance with MIL-DTL-16377 applied as shown on figures 1 and 3:

a. Two Type I.
b. One Type II for Symbols 331.2 and 347.3 only.
c. One Type IV. The information to be shown on the Type IV label shall be "USE THE FOLLOWING REPAIR PARTS ONLY:" followed by a list of all standard parts shown in table II and window assemblies of table VI by name and stock number only.

Standard parts: The standard parts shown in table II shall be used in the construction of the fixtures specified herein.

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SSL REQUIREMENTS:

CLASSIFICATION:

Type: III
Class: 2

SYMBOL NUMBERS:

331.1SSL: Complete lighting fixture assembly, standard mounting, with LED light source and white diffusing window assembly.
331.2SSL: Complete lighting fixture assembly, flush mounting, with LED light source and white diffusing window
347.2SSL: Complete lighting fixture assembly, standard mounting, with LED light source and clear and white prismatic window.
347.3SSL: Complete lighting fixture assembly, flush mounting, with LED light source and clear and white prismatic window.

DESIGN:

Electrical rating:	20 watts maximum, 120 volts, 60 hertz.
Dimensions and configuration:	External dimensions cannot exceed those specified on figure 1.
Photometric data:	See figure 4 and 3.7.1 of MIL-DTL-16377.
Chromaticity:	See 3.7.4 of MIL-DTL-16377.
Fail-safe circuit:	See 3.17 of MIL-DTL-16377.
Wiring:	Must maintain same control scheme for light output as legacy fluorescent fixture. Refer to figure 22 of MIL-DTL-16377.
Mounting dimensions:	
Symbols 331.1SSL and 347.2SSL:	See figure 25 of MIL-DTL-16377.
Symbols 331.2SSL and 347.3SSL:	See figure 26 of MIL-DTL-16377.
Window assembly:	
Symbols 331.1SSL and 331.2SSL:	White diffuse.
Symbols 347.2SSL and 347.3SSL:	Clear and white prismatic.
Enclosure:	Watertight. See 3.5.11 of MIL-DTL-16377.
Cable:	10 feet of Type LSDHOF-3 in accordance with MIL-DTL-24643/3.
Weight:	
Symbols 331.1SSL and 347.2SSL:	8 pounds 10 ounces maximum.
Symbols 331.2SSL and 347.3SSL:	10 pounds 2 ounces maximum.
Designation and marking:	The following labels shall be furnished in accordance with MIL-DTL-16377 applied as shown on figures 1 and 3:

a. Two Type I.
b. One Type II for Symbols 331.2SSL and 347.3SSL only.
c. One Type IV. The information to be shown on the Type IV label shall be “USE THE FOLLOWING REPAIR PARTS ONLY:” followed by:
   (1) A list of all standard parts shown in table II and window assemblies of table VI by name and stock number only, if appropriate, or
   (2) A list of available replacement parts and stock numbers, or if unavailable, manufacturer’s part numbers.

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EXAMINATION AND TESTS:

Examination and tests shall be as specified in MIL-DTL-16377 and as follows:

First article inspection:
Complete fixtures: See table III for Type I and table IV for Type III.
Window assemblies: See table V.
Conformance:
Comparison inspection: Same as first article inspection.
Conformance inspection: See MIL-DTL-16377.

GENERAL INFORMATION:

Assignment of military part numbers and general information is shown in table VI.

TABLE II. Standard parts for type I fixtures.

Name	Specification or Military Part Number	National Stock Number (NSN)	Remarks
Lamp	A-A-55005-1	6240-00-152-2996	Industry number F20T12/CW
Lamp lock	M16377/40-002	6250-00-399-7040	
Lampholder	UL 496	6250-00-284-0481	SUPP. SB, FIG. SB12
Startersocket	UL 496	6250-00-295-2738	SUPP. SB, FIG. SB14
Starter	NEMA ANSI C78.180	6250-00-299-2884	Commercial designation: FS2
Ballast	M16377/72-001	6250-01-465-6716	
Shock mount	M16377/41-009 or M16377/41-010	6210-00-973-1747 6210-00-973-1748	See 3.5.16 of MIL-DTL-16377
Red filter tube	M16377/42-003	6210-00-295-2838	Not furnished, see 3.7.9 of MIL-DTL-16377
Broad band blue filter tube	M16377/42-005	6210-00-441-5138	Not furnished, see 3.7.9 of MIL-DTL-16377
Amber filter tube	M16377/42-006	6210-01-166-7184	Not furnished, see 3.7.9 of MIL-DTL-16377
Gray filter tube (1% transmission)	M16377/42-007	6210-01-498-8001	Not furnished, see 3.7.9 of MIL-DTL-16377
Gray filter tube (2.5% transmission)	M16377/42-009	6210-01-313-8168	Not furnished, see 3.7.9 of MIL-DTL-16377

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TABLE III. First article and comparison inspection for type I fixtures. [1/]

Test	Requirement	Test Method	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Power interface	3.6.18	4.8.21	
Power quality	3.6.18.1	4.8.22	
Voltage transient spike	3.6.18.2	4.8.23	
Electromagnetic interference	3.6.14	4.8.16	
Dielectric withstanding voltage	3.6.15	4.8.2	2/
Insulation resistance	3.6.16	4.8.3	2/
Enclosure effectiveness	3.5.11	4.8.14.4	2/, 3/
Shock	3.5.12	4.8.7	2/
Vibration	3.5.13	4.8.8	2/
Enclosure effectiveness	3.5.11	4.8.14.4	2/, 3/
Dielectric withstanding voltage	3.6.15	4.8.2	2/
Insulation resistance	3.6.16	4.8.3	2/
Noise	3.5.14	4.8.9	2/
Continuity of grounding	3.6.13.4	4.8.12	
Optical uniformity	3.7.7.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Light output	3.7.1	4.8.6.1	Candlepower
Magnetic permeability	3.4.8	4.8.18	

NOTES:

1/ Requirement and test method paragraph numbers refer to MIL-DTL-16377.

2/ In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.

3/ Watertight.

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TABLE IV. First article and comparison inspection for type III luminaires. 1/

Test	Requirement	Test Method	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Power interface	3.6.18	4.8.21	
Power quality	3.6.18.1	4.8.22	
Voltage transient spike	3.6.18.2	4.8.23	
Electromagnetic interference	3.6.14	4.8.16	
Dielectric withstanding voltage	3.6.15	4.8.2	2/
Insulation resistance	3.6.16	4.8.3	2/
Enclosure effectiveness	3.5.11	4.8.14	2/, 3/
Shock	3.5.12	4.8.7	2/
Vibration	3.5.13	4.8.8	2/
Enclosure effectiveness	3.5.11	4.8.14	2/, 3/
Dielectric withstanding voltage	3.6.15	4.8.2	2/
Insulation resistance	3.6.14	4.8.3	2/
Noise	3.5.14	4.8.9	2/
Continuity of grounding	3.6.13.4	4.8.12	
Photometrics	3.7.1	4.8.6	Candlepower
Brightness	3.7.5	4.8.6.4	
Chromaticity	3.7.4	4.8.6.5	
Correlated color temperature (CCT)	3.7.4.1.1	4.8.6.6	
Lumen maintenance	3.7.11	4.8.20	
Fail-safe circuit design	3.17	4.8.24	
Optical uniformity	3.7.7.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Magnetic permeability	3.4.8	4.8.18	

NOTES:

1/ Requirement and test method paragraph numbers refer to MIL-DTL-16377.

2/ In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.

3/ Watertight.

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TABLE V. First article and comparison inspection for window assemblies. [1/]

Tests	Requirement	Test Method	Remarks
Examination	3.13	4.5	
Optical uniformity	3.7.7.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Light output	3.7.1	4.8.6.1	Candlepower

NOTE:
[1/] Requirements and test method paragraph numbers refer to MIL-DTL-16377.

TABLE VI. Part number and general information.

Military Part [1/] Number M16377/8-	Item Name	National Stock Number (NSN)	Remarks
331.1	Complete fixture, Symbol 331.1	6210-00-548-0598	[2/]
331.2	Complete fixture, Symbol 331.2	6210-01-088-2511	[2/]
347.2	Complete fixture, Symbol 347.2	6210-01-088-6945	
347.3	Complete fixture, Symbol 347.3	6210-01-088-6186	
331.1SSL	Complete fixture, solid state, Symbol 331.1SSL		New
331.2SSL	Complete fixture, solid state, Symbol 331.2SSL		New
347.2SSL	Complete fixture, solid state, Symbol 347.2SSL		New
347.3SSL	Complete fixture, solid state, Symbol 347.3SSL		New
001	Window, white diffusing	6210-00-548-0601	[3/]
003	Window, clear and white prismatic	6210-01-438-2960	

NOTES:
[1/] Military part numbers M16377/8-347, 347.1, and 002 have been superseded by military part numbers M16377/8-347.2, 347.3, and M16377/67-001, respectively.
[2/] Formerly covered on Drawing NAVSHIPS 9000-S6401-74482 and MS17194.
[3/] Formerly covered on MS16891-40.

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CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Navy – SH

Review Activity:
DLA – IS

Preparing activity:
Navy – SH
(Project 6210-2012-004)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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